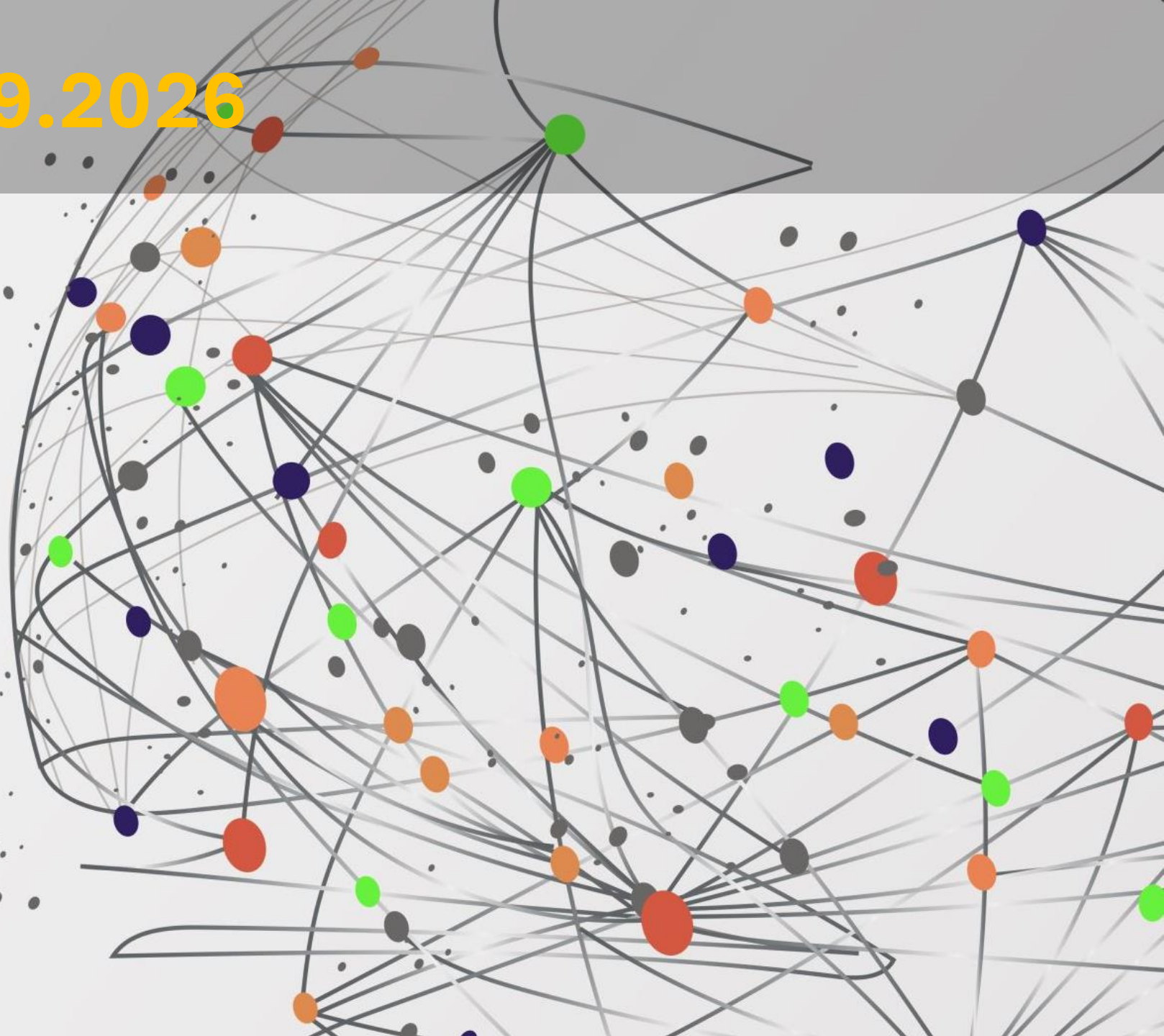
Next steps 21.09.2026



Mandatory Assignment

Running Example

Work with AI plus survey

Mandatory Assignment – 15.11.2026

Consists of two parts:

PART I – Oral presentation of the design process done for the development of the desktop app (due date 09.11.2026 and 11.11.2026)

PART II – Programming in ProTuS - each student needs to complete 35 challenges and 25 coding exercises in ProTuS (due date 15.11.2026)

Oral presentation – 09 and 11 Nov 2026

Each group has to prepare a presentation for the design part of your desktop presentation. The presentation needs to contain:

1. Introduction of the team
2. Introduction of the problem (for your customer) and your proposed solution (MVP)
3. Requirements analysis (user stories, FR, NFR that align with the need you defined from your client)
4. Use cases and use case diagram (must have include and extend relationships)
5. Design class diagram (classes need to contain examples of constructor, interface and implementation methods).
6. Sequence diagram for one selected use case (complex enough to contain several classes)
7. If you have mock ups or already programmed part of your app – Optional but advisable.

Programming in ProTuS – 15.11.2026

Each student need to register in ProTuS following the instruction document that will be uploaded in Canvas. Once you register, you will need to complete 35 challenges and 25 coding exercises.

We do not expect you to do that in one or two days, but from today till 15.11.2026. Thus, we need you to use ReguLA – where you will self-regulate your time, effort, learning, and completion of tasks.

Self-regulated learning is especially important now, when conversational AI is overtaking education and learning processes.

- What is ReguLA? - is designed to help you: **plan** your study work, **monitor** your progress, and **reflect** on what works (and what does not) for you. We will upload guidelines how to use ReguLA.

Running Example

Modern cinemas require a flexible and efficient software system to manage daily operations, including movie scheduling, customer reservations, ticket reservations, customer information, and operational reporting. The cinema chain operates several branches, each containing one or more cinema halls with different seating capacities. Movies may be shown at different branches and at different times, and the available schedule changes regularly as movies are added or removed from the catalogue.

Cinema staff need a single system through which they can register customers, search upcoming screenings, make or cancel reservations, select available seats, and record ticket sales. Managers additionally need to maintain the movie catalogue and screening schedule and access information about attendance, reservations, sold-out screenings, and revenue.

The system must maintain accurate information about movies, halls, screenings, customers, reservations, seats, and ticket sales. It must prevent conflicting screenings in the same hall and prevent the same seat from being reserved or sold to more than one customer for the same screening.

Work with AI and finish the survey after

Use ChatGPT to work on the design of the Cinema Manager desktop app

https://chatgpt.com/g/g-p-6aaff1abdf0481918cf045777adaaa52-running-example-oap-fall-26/project

You should put yourself into the position of software consultant that is talking to the manager of Cinema Inc. Ask questions to understand the need, the issues/problems the manager has. Ask questions while performing requirements analysis so that you can write functional and non-functional requirements. Continue by asking questions while you work on the use cases and draw the use-case diagram. Then, ask questions while you work on the design class diagram for the whole app and the sequence diagram for one use case of your own choosing. Once you are done, tell ChatGPT that you are done, so that you can get feedback on your work.

Survey and a Diary after you are done with the Running Example

Initial survey https://luisa-ruhland.limesurvey.net/students-initial?lang=en&newtest=Y

Diary https://luisa-ruhland.limesurvey.net/students-microdiary?lang=en&newtest=Y

Please enter the Course ID provided by your instructor/lecturer: **OAP2026**

FINAL EXAM - Private Book Library Manager

Many people own books that are spread across different rooms, bookshelves, boxes, or storage locations. Over time, it can become difficult to remember which books are already owned, where they are placed, which books have been read, and which books belong to a certain writer or genre. A lightweight desktop application can help private users organize and search their personal book collection in a structured way.

The user wants to develop a desktop application for managing a private book library. The system should allow the user to register books, writers, genres, and bookshelf locations. It should also support tracking whether a book has been read or not, and make it easy to search and review the collection. The application should provide a clear overview of the books owned by the user, including where each book is located and which books belong to specific writers or genres.

FINAL EXAM - Private Book Library Manager

System Goals

1. Maintain a registry of books in a private collection.
2. Store information about writers, genres, and bookshelf locations.
3. Track whether a book has been read or not.
4. Provide quick access to information about owned books.
5. Support searching and filtering by book title, writer, genre, bookshelf, and reading status.
6. Generate simple reports about the private library collection.